

RantLab: 24-Hour Task Sheet

Team Fantastic 4 — Open Innovation Track (Online Demo) | Free open-source stack (FastAPI, Next.js, Ollama, Whisper)

1. Team Overview & Owned Files

Member & Role	Key Responsibilities & Files Owned
Akilan <i>Team Lead</i>	Pipeline, LLM prompt, integration backend/app/main.py, llm.py, models.py, config.py, docs/api-contract.md
Arvind <i>AI & DB</i>	Speech to text, database, embeddings, clustering backend/app/stt.py, db.py, embed.py, cluster.py
Jayasree <i>Frontend & Demo Lead</i>	Frontend: recorder UI, card view, share page, board (Drives live online demo) frontend/ (lib/api.ts, lib/types.ts, mocks/, app pages)
Priyan <i>QA & Demo Deck</i>	Test data, evaluation rubric, bug log, deck & backup video data/, scripts/, backend/tests/, README.md

2. 24-Hour Phase & Task Timeline

Setup (H0 to H1) — *Checkpoint H1: Everyone can run the repo and their own tool locally.*

Member	Task	Detail	Time	Type
Akilan	Create repo & API contract	Free GitHub repo, api-contract.md with JSON shape.	0.5h	MVP
Akilan	Install Ollama & pull qwen2.5:3b	Test response time under 15s on local machine.	0.5h	MVP
Arvind	Install faster-whisper & ffmpeg	Transcribe test clip on CPU. Pick tiny/base model.	0.5h	MVP
Jayasree	Scaffold Next.js, TS & Tailwind	Setup project, code against mock JSON.	0.5h	MVP
Priyan	Write 40 campus complaints	Save as seed.json with near-duplicates.	1.0h	MVP

Build the parts (H1 to H6) — *Checkpoint H6: A typed complaint returns three fixes from the API.*

Member	Task	Detail	Time	Type
Akilan	Write fix-generation prompt	Ollama JSON mode, pydantic validation, retry once.	3.0h	MVP
Akilan	FastAPI app POST /api/rant/text	Text input in, card JSON output out with CORS.	1.5h	MVP
Arvind	Write transcribe(audio)	Accept webm/ogg audio, convert with faster-whisper.	2.0h	MVP
Arvind	SQLite schema	Cards, complaints, embeddings tables (anonymous).	1.5h	MVP
Arvind	Embedding helper	Load MiniLM at startup, embed(text) vector helper.	1.5h	MVP
Jayasree	Record screen UI	MediaRecorder, 30s cap, timer, typed option.	3.0h	MVP
Jayasree	Concept card component	Problem, pattern, 3 fix rows (cost/timeframe).	2.0h	MVP
Priyan	Record 6 voice samples	Different voices, noisy clip for Whisper test.	1.5h	MVP
Priyan	Evaluation rubric	Score relevance, specificity, sensible owner.	1.5h	MVP

Wire it together (H6 to H12) — *Checkpoint H12: Voice in the browser produces a card.*

Member	Task	Detail	Time	Type
Akilan	Full pipeline POST /api/rant	Audio/text -> transcribe -> LLM -> save DB -> card.	3.0h	MVP
Akilan	Set up HTTPS tunnel	Cloudflare Tunnel / ngrok for online mic access.	1.0h	MVP
Arvind	Clustering	Cosine similarity grouping for Wi-Fi / canteen rants.	3.0h	MVP
Jayasree	Connect to real API	Audio upload, loading stages (Transcribing/Thinking).	2.0h	MVP
Jayasree	Card page at /card/[id]	Public link for online demo screen sharing.	2.0h	MVP
Priyan	Test model on 10 complaints	Prompt evaluation as soon as prompt is ready.	3.0h	MVP
Priyan	README & architecture diagram	Setup steps and flow diagram for online presentation.	2.0h	MVP

MVP Freeze & Testing (H12 to H16) — *Checkpoint H16: Feature freeze. After this, only bug fixes.*

Member	Task	Detail	Time	Type
Akilan	Merge & fix top bugs	Merge branches, run end-to-end QA fixes.	2.0h	MVP
Akilan	Tag v1-mvp	Tag working build on Git for Jayasree's demo machine.	0.25h	MVP
Arvind	GET /api/board	Return clusters with counts, labels, complaints.	1.5h	MVP
Arvind	Speed check	Benchmark upload, STT, and LLM steps.	1.5h	MVP
Jayasree	Copy link / PNG download	html-to-image package export.	1.5h	Stretch
Jayasree	Mobile & browser pass	Safari/Chrome mic permissions check.	1.5h	MVP
Priyan	QA pass & bug log	Stress test text, voice, online network conditions.	3.0h	MVP
Priyan	Privacy checklist	Verify anonymous storage, short notice.	1.0h	MVP

Stretch & Polish (H16 to H20) — *Checkpoint H20: Stop building. Nothing new after this.*

Member	Task	Detail	Time	Type
Akilan	Tune prompt on weak outputs	Re-run test set, refine vague fixes & owners.	2.0h	MVP
Akilan	Model fallback switch	Environment flag to switch to cloud API if needed.	1.0h	Stretch
Arvind	Handle messy audio	Silence, noise, accents -> clear error message.	2.0h	MVP
Arvind	Rank recurring issues	Sort clusters by recency and count.	1.0h	Stretch
Jayasree	Board page at /board	List clusters with counts and complaints.	3.0h	Stretch
Jayasree	Polish states & branding	Loading, empty states, logo, tagline.	1.0h	MVP
Priyan	Backup demo video	90s OBS recording for screen-share fallback.	2.0h	MVP
Priyan	Visual concept for top fix	Generate optional fix preview image.	2.0h	Stretch

Pitch Prep & Buffer (H20 to H24) — *Rehearsals: Two full online 8-minute run-throughs, Jayasree driving demo screen.*

Member	Task	Detail	Time	Type
Akilan	Demo environment & team rehearsal	Support Jayasree with backend setup, test slides 1-2.	3.0h	MVP
Arvind	Speaker 2 rehearsal & Tech Q&A;	Arvind rehearses slides 3-4 (Whisper, MiniLM).	2.5h	MVP
Jayasree	Online Demo Lead & Speaker 3	Jayasree drives screen-share live demo under 60s.	2.5h	MVP
Priyan	Real numbers into slides & Speaker 4	Priyan updates slide 7 metrics, team Q&A; sheet.	3.0h	MVP

3. Individual Member Worksheets & Complete Task Lists

Comprehensive task list for each person across the entire 24-hour hackathon duration.

Subtopic 3.1: Akilan — Team Lead: Backend, LLM Prompt & Integration

Files Owned: backend/app/main.py, llm.py, models.py, config.py, docs/api-contract.md

Overview: Responsible for overall architecture, API endpoints, Ollama LLM prompt engineering, response validation, and backend integration.

Phase	Task Name	Detailed Instructions	Hours	Priority
Setup	Create repo & API contract	Free GitHub repo, api-contract.md with JSON shape.	0.5h	MVP
Setup	Install Ollama & pull qwen2.5:3b	Test response time under 15s on local machine.	0.5h	MVP
Build the parts	Write fix-generation prompt	Ollama JSON mode, pydantic validation, retry once.	3.0h	MVP
Build the parts	FastAPI app POST /api/rant/text	Text input in, card JSON output out with CORS.	1.5h	MVP
Wire it together	Full pipeline POST /api/rant	Audio/text -> transcribe -> LLM -> save DB -> card.	3.0h	MVP
Wire it together	Set up HTTPS tunnel	Cloudflare Tunnel / ngrok for online mic access.	1.0h	MVP
MVP Freeze & Testing	Merge & fix top bugs	Merge branches, run end-to-end QA fixes.	2.0h	MVP
MVP Freeze & Testing	Tag v1-mvp	Tag working build on Git for Jayasree's demo machine.	0.25h	MVP
Stretch & Polish	Tune prompt on weak outputs	Re-run test set, refine vague fixes & owners.	2.0h	MVP
Stretch & Polish	Model fallback switch	Environment flag to switch to cloud API if needed.	1.0h	Stretch
Pitch Prep & Buffer	Demo environment & team rehearsal	Support Jayasree with backend setup, test slides 1-2.	3.0h	MVP

Total Allocated Hours: 17.75 hrs

Subtopic 3.2: Arvind — AI & DB Specialist: Speech, Embeddings & Clustering

Files Owned: backend/app/stt.py, db.py, embed.py, cluster.py

Overview: Responsible for local Speech-to-Text via faster-whisper, SQLite database storage, MiniLM vector embeddings, and complaint clustering.

Phase	Task Name	Detailed Instructions	Hours	Priority
Setup	Install faster-whisper & ffmpeg	Transcribe test clip on CPU. Pick tiny/base model.	0.5h	MVP
Build the parts	Write transcribe(audio)	Accept webm/ogg audio, convert with faster-whisper.	2.0h	MVP
Build the parts	SQLite schema	Cards, complaints, embeddings tables (anonymous).	1.5h	MVP
Build the parts	Embedding helper	Load MiniLM at startup, embed(text) vector helper.	1.5h	MVP
Wire it together	Clustering	Cosine similarity grouping for Wi-Fi / canteen rants.	3.0h	MVP
MVP Freeze & Testing	GET /api/board	Return clusters with counts, labels, complaints.	1.5h	MVP
MVP Freeze & Testing	Speed check	Benchmark upload, STT, and LLM steps.	1.5h	MVP
Stretch & Polish	Handle messy audio	Silence, noise, accents -> clear error message.	2.0h	MVP
Stretch & Polish	Rank recurring issues	Sort clusters by recency and count.	1.0h	Stretch
Pitch Prep & Buffer	Speaker 2 rehearsal & Tech Q&A;	Arvind rehearses slides 3-4 (Whisper, MiniLM).	2.5h	MVP

Total Allocated Hours: 17.00 hrs

Subtopic 3.3: **Jayasree** — Frontend Lead & Online Demo Presenter

Files Owned: frontend/ (lib/api.ts, lib/types.ts, mocks/, app pages)

Overview: Responsible for Next.js web UI, audio recording interface, shareable card view, board UI, and driving the live screen-shared demo during online judging.

Phase	Task Name	Detailed Instructions	Hours	Priority
Setup	Scaffold Next.js, TS & Tailwind	Setup project, code against mock JSON.	0.5h	MVP
Build the parts	Record screen UI	MediaRecorder, 30s cap, timer, typed option.	3.0h	MVP
Build the parts	Concept card component	Problem, pattern, 3 fix rows (cost/timeframe).	2.0h	MVP
Wire it together	Connect to real API	Audio upload, loading stages (Transcribing/Thinking).	2.0h	MVP
Wire it together	Card page at /card/[id]	Public link for online demo screen sharing.	2.0h	MVP
MVP Freeze & Testing	Copy link / PNG download	html-to-image package export.	1.5h	Stretch
MVP Freeze & Testing	Mobile & browser pass	Safari/Chrome mic permissions check.	1.5h	MVP
Stretch & Polish	Board page at /board	List clusters with counts and complaints.	3.0h	Stretch
Stretch & Polish	Polish states & branding	Loading, empty states, logo, tagline.	1.0h	MVP
Pitch Prep & Buffer	Online Demo Lead & Speaker 3	Jayasree drives screen-share live demo under 60s.	2.5h	MVP

Total Allocated Hours: 19.00 hrs

Subtopic 3.4: Priyan — QA, Test Data, Privacy & Demo Deck Lead

Files Owned: data/, scripts/, backend/tests/, README.md

Overview: Responsible for seed dataset creation, prompt evaluation rubric, QA bug logging, backup demo video, slide deck preparation, and tech Q&A.;

Phase	Task Name	Detailed Instructions	Hours	Priority
Setup	Write 40 campus complaints	Save as seed.json with near-duplicates.	1.0h	MVP
Build the parts	Record 6 voice samples	Different voices, noisy clip for Whisper test.	1.5h	MVP
Build the parts	Evaluation rubric	Score relevance, specificity, sensible owner.	1.5h	MVP
Wire it together	Test model on 10 complaints	Prompt evaluation as soon as prompt is ready.	3.0h	MVP
Wire it together	README & architecture diagram	Setup steps and flow diagram for online presentation.	2.0h	MVP
MVP Freeze & Testing	QA pass & bug log	Stress test text, voice, online network conditions.	3.0h	MVP
MVP Freeze & Testing	Privacy checklist	Verify anonymous storage, short notice.	1.0h	MVP
Stretch & Polish	Backup demo video	90s OBS recording for screen-share fallback.	2.0h	MVP
Stretch & Polish	Visual concept for top fix	Generate optional fix preview image.	2.0h	Stretch
Pitch Prep & Buffer	Real numbers into slides & Speaker 4	Priyan updates slide 7 metrics, team Q&A; sheet.	3.0h	MVP

Total Allocated Hours: 20.00 hrs